Using Dilations to Reason About Similarity
Answer Key

Grade 8 / Pre-Algebra — Geometry

Quick Answers

1. 6, 10	4. 1.5, 1.5	7. 7, 4	10. 3.5, 28
2. 4, -3	5. —	8. 1.5, 12	
3. 10, 15, 1.5	6. 2.5, 10	9. 12, 48, 4	

Common wrong answers

3. *If they got 0.67: divided the original length by the image length instead of image over original*
7. *If they got 9: dilated as if the center were the origin, ignoring the center at (1, 1)*
10. *If they got 2.29: used the ratio upside down, scaling by 6 over 21 instead of 21 over 6*

1. $P(3,5)$**, centre the origin,** $k = 2$**.** With the centre at the origin the rule is $(x,y) \mapsto (kx,ky)$: each coordinate is the distance from an axis, and every distance from the centre is multiplied by k. x-coordinate: $2 \times 3 = \boxed{6}$.
y-coordinate: $2 \times 5 = \boxed{10}$.
So P' is the point $(6,10)$, twice as far from the origin as P in the same direction.

2. $Q(8,-6)$**, centre the origin,** $k = 0.5$**.** Same rule, $(x,y) \mapsto (kx,ky)$, and a negative coordinate is halved just like a positive one. x-coordinate: $0.5 \times 8 = \boxed{4}$.
y-coordinate: $0.5 \times (-6) = \boxed{-3}$.
So Q' is $(4,-3)$. Because $0 < k < 1$, the image is a *reduction*: every length is half of the original.

3. $\overline{AB}$ **from** $(0,0)$ **to** $(6,8)$**; image from** $(0,0)$ **to** $(9,12)$**.** (a) Distance formula: $AB = \sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100}$, so $AB = \boxed{10 \text{ units}}$.
(b) $A'B' = \sqrt{9^2+12^2} = \sqrt{81+144} = \sqrt{225}$, so $A'B' = \boxed{15 \text{ units}}$.
(c) The scale factor is image length divided by original length — that order, because k answers "how many times as long is the image?"

$$k = \frac{15}{10} = 1.5$$

Scale factor: $\boxed{1.5}$. Check: $1.5 \times 6 = 9$ and $1.5 \times 8 = 12$, which are exactly the image coordinates.

4. Slope of the ramp and of its image. Original, from $(2, 3)$ to $(6, 9)$:

$$\text{slope} = \frac{9-3}{6-2} = \frac{6}{4} = 1.5$$

so the original slope is $\boxed{1.5}$.
Image, from $(6, 9)$ to $(18, 27)$:

$$\text{slope} = \frac{27-9}{18-6} = \frac{18}{12} = 1.5$$

so the image slope is $\boxed{1.5}$.
The two slopes are equal, so the image segment is parallel to the original (it lies on the same line here, since the centre is on that line's ray). Equal slopes are the coordinate evidence that a dilation changes size without turning the figure — which is why corresponding angles are unchanged and the figures are similar.

5. Why $\triangle A'B'C'$ is similar to $\triangle ABC$ (open reasoning — grade against this model). A dilation about the origin with $k = 3$ sends each vertex (x, y) to $(3x, 3y)$, so every side length of the triangle is multiplied by the same factor 3: all three ratios $A'B'/AB$, $B'C'/BC$ and $A'C'/AC$ equal 3. A dilation also maps each side to a segment with the same slope, so each side of the image is parallel to the matching side of the original and the three angle measures are unchanged. Corresponding angles congruent *and* corresponding sides in one constant ratio is exactly the definition of similar figures, so $\triangle A'B'C' \sim \triangle ABC$ with ratio 3. Accept any answer that names both facts: sides all scaled by the same k, angles unchanged.

6. Photograph 4 cm by 6 cm, the 6 cm side becomes 15 cm. (a) The scale factor is the new length over the matching old length:

$$k = \frac{15}{6} = 2.5$$

Scale factor: $\boxed{2.5}$.
(b) Similar figures scale *both* sides by the same factor, so the short side becomes $2.5 \times 4 = 10$. Enlarged short side: $\boxed{10\text{ cm}}$. Check the shape is unchanged: $10 : 15$ reduces to $2 : 3$, the same ratio as $4 : 6$.

7. Centre $C(1, 1)$, $k = 3$, point $M(3, 2)$. When the centre is not the origin, measure from the centre, scale, then step back out from the centre: $(x, y) \mapsto \big(h + k(x-h),\ m + k(y-m)\big)$ with $h = m = 1$. x-coordinate: $1 + 3(3-1) = 1 + 6 = \boxed{7}$.
y-coordinate: $1 + 3(2-1) = 1 + 3 = \boxed{4}$.
So M' is $(7, 4)$. Note $CM' = 3\,CM$, and M' is on ray CM — a common error is to multiply the coordinates by 3 and land on $(9, 6)$, which dilates about the origin instead.

8. Model sail 6 cm and 8 cm; the larger sail's matching side is 9 cm. (a) The two sails are similar, so one factor takes every model length to the matching larger length:

$$k = \frac{9}{6} = 1.5$$

Scale factor: $\boxed{1.5}$.
(b) Apply the same factor to the other side: $1.5 \times 8 = 12$. Matching side: $\boxed{12 \text{ cm}}$. Check the ratios agree: $9 : 12$ reduces to $3 : 4$, the same as $6 : 8$.

9. Rectangle $(0, 0), (4, 0), (4, 3), (0, 3)$ **dilated by** $k = 2$**.** (a) The original is 4 units by 3 units, so its area is $4 \times 3 = 12$. Original area: $\boxed{12 \text{ square units}}$.
(b) The image is 8 units by 6 units, so its area is $8 \times 6 = 48$. Image area: $\boxed{48 \text{ square units}}$.
(c) Dividing, $48 \div 12 = 4$, so the area ratio is $\boxed{4}$. That number is k^2: $2^2 = 4$. Both dimensions were multiplied by k, so the area was multiplied by k twice. Lengths scale by k; areas scale by k^2.

10. Scale drawing: flagpole 8 cm, wall 6 cm; the real wall is 21 m. The drawing and the yard are similar, so the same factor takes every drawing length to the matching real length. Use the pair given in both figures — the wall:

$$k = \frac{21 \text{ m}}{6 \text{ cm}} = 3.5 \text{ m per cm}$$

Scale factor: $\boxed{3.5}$.
Now apply that factor to the flagpole's drawing height: $8 \times 3.5 = 28$. Real height of the flagpole: $\boxed{28 \text{ m}}$. Check: $28 : 21$ reduces to $4 : 3$, and $8 : 6$ also reduces to $4 : 3$, so the drawing and the yard really are similar.